

Chapter 3: Atomic Structure

Atoms

- Atoms are the basic building blocks of matter.
- They are the smallest possible particle of an element that can take part in a chemical change.
- An atom is made up of 3 different ^{sub-atomic} particles.
→ protons, neutrons, electrons

Proton

Symbol → p

Relative Mass → 1

Relative Charge → 1+

Neutron

Symbol → n

Relative Mass → 1

Relative Charge → 0

Electron

Symbol → e

Relative Mass → $\frac{1}{1840}$

Relative Charge → 1-

Structure of an atom

- The nucleus is in the centre of an atom, made of protons and neutrons.
- The electrons move around the nucleus at great speed.

Properties of atoms

- The mass of electrons is negligible compared to the mass of protons and neutrons. → Mass is calculated by protons + neutrons
- Most of the atom is empty space.
- All atoms are electrically neutral.
↳ Number of protons = number of electrons

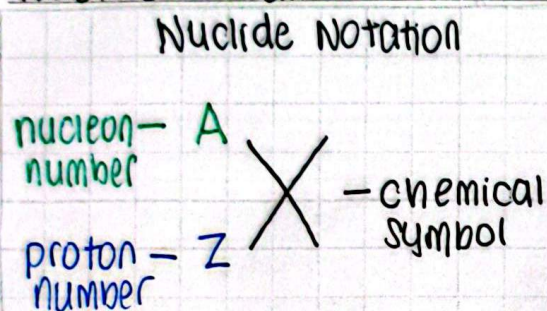
Proton number (aka Atomic number)

- Proton number, ~~Z~~, is the number of protons in an atom.
- ★ Each element in the periodic table has a unique proton number.
- Symbol: Z

~~Nea~~ Nucleon number (aka Mass number)

- Nucleon number is the number of protons and neutrons in the nucleus of an atom.
- $\therefore$ Number of neutrons = nucleon number - proton number
- Symbol: A, or relative atomic mass (A_r)

Nuclide Notation



Example: $^{16}_8\text{O}$

Nucleon number = 16

Proton number = 8

Number of neutrons = $16 - 8$

$= 8$

vs Periodic Table:

Key	
(Z) proton (atomic) number	11 $\rightarrow$ no. of protons
atomic symbol	Na
name	sodium
(A) relative atomic mass	23 $\rightarrow$ no. of protons + no. of neutrons

Isotopes

- Isotopes are atoms of the same element with the same number of protons but different number of neutrons.
- The atomic mass of an element is the average of the nucleon numbers of its ~~two~~ isotopes, according to their relative abundance.

Example: Chlorine element

Chlorine-35 isotope $\rightarrow$ relative abundance of 75%

Chlorine-37 isotope $\rightarrow$ relative abundance of 25%

$$\therefore \text{Relative atomic mass of chlorine} = 0.75 \times 35 + 0.25 \times 37 = 35.5$$

- Isotopes have different masses because they have different number of neutrons.

- Isotopes have ~~the~~ similar chemical properties but have slightly different physical properties.

Electronic configuration of an atom

- For the first 20 elements, the maximum number of electrons in the first 3 shells are:

1st electron shell $\rightarrow 2$

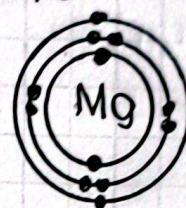
2nd electron shell $\rightarrow 8$

3rd electron shell $\rightarrow 8$

Drawing electron shells:

- Follow the algorithm NSEW.

Example



$\rightarrow$ Electrons

can be represented by x or •

- The electronic configuration shows the arrangement of electrons in an atom.

How to write:

Format: a, b, c (each representing the number of electrons in that shell)

Examples: 2, 8, 2, 2, 7, 2, 2

Valence electron shell

- The valence electron shell is the outermost electron shell that is the furthest away from the nucleus.

- Electrons in the valence electron shell are called valence electrons.

- The number of valence electrons in an atom determines the chemical properties of the atom.

- Same number of valence electrons = similar chemical property.

Cheatsheet: No. of electrons in valence shell from periodic table

Group 1: 1

Group 2: 2

Group 13: 3

Group 14: 4

Group 15: 5

Group 16: 6

Group 17: 7

Group 18: 8

- The horizontal rows of elements are called periods.
- Periods show the number of electron shells of the atom.

Chapter 5.1 Structure & Properties of Materials: Elements, Compounds, Mixtures

Elements

- An element is a substance which cannot be broken down into two or more simpler substances by chemical means.
- An element consists of only one type of atom.

Metalloids

- Metalloids are elements with both metallic and non-metallic properties.
- They are elements along the zig-zag line, e.g. ~~Boron~~ B, Si, Ge, As, Sb, Te.

Properties of metals vs non metals

Metals	Non-metals
- Shiny appearance	- Dull
- Solids at room temperature, high melting and boiling points (except mercury, sodium, potassium)	- Gases or liquids at room temperature, low melting and boiling points (except carbon and silicon)
- Malleable (can be hammered into shape), Ductile (can be stretched into wires)	- Brittle (if solid)
- Good conductors of heat & electricity	- Poor conductors of heat & electricity

Atoms & molecules of elements

- Elements can exist as monatomic elements (individual atoms) or molecules of elements (two or more atoms chemically combined)

Elements that exist naturally as molecules

1. diatomic molecule (2 atoms) $\rightarrow$ H_2 , N_2 , O_2 , F_2 , Cl_2 , Br_2 , I_2
2. triatomic molecule (3 atoms) [Have no fear of Ice cold Beer]
3. polyatomic molecule (>3 atoms)

Compounds

- A compound is a pure substance containing two or more elements that are chemically combined in a fixed ratio.
- Atoms of different elements form molecules of compounds.
- A compound has different properties from the elements that form it.

- The constituent elements in a compound have a fixed proportion by mass.

Example: $\text{H}_2\text{O} \rightarrow \text{H}:\text{O} \rightarrow 2:1$

- Compounds are represented by their chemical formula.

- Examples: Sodium chloride = sodium + chlorine



Mixtures

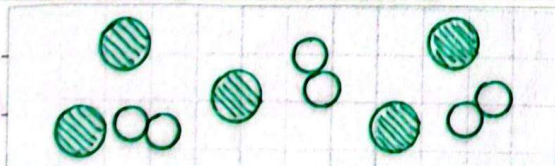
- A mixture contains two or more substances physically mixed together.

- Mixtures do not have fixed compositions.

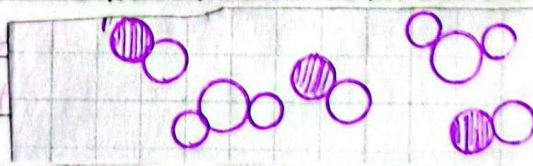
- Mixtures can be separated by physical means e.g. filtration.

Types of mixtures

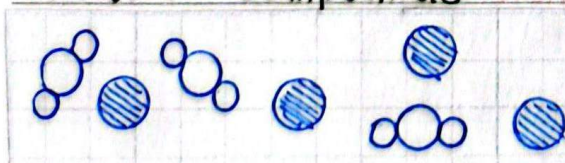
Elements + Elements



Compounds + Compounds



Elements + compounds



Melting point and Boiling point of Mixtures

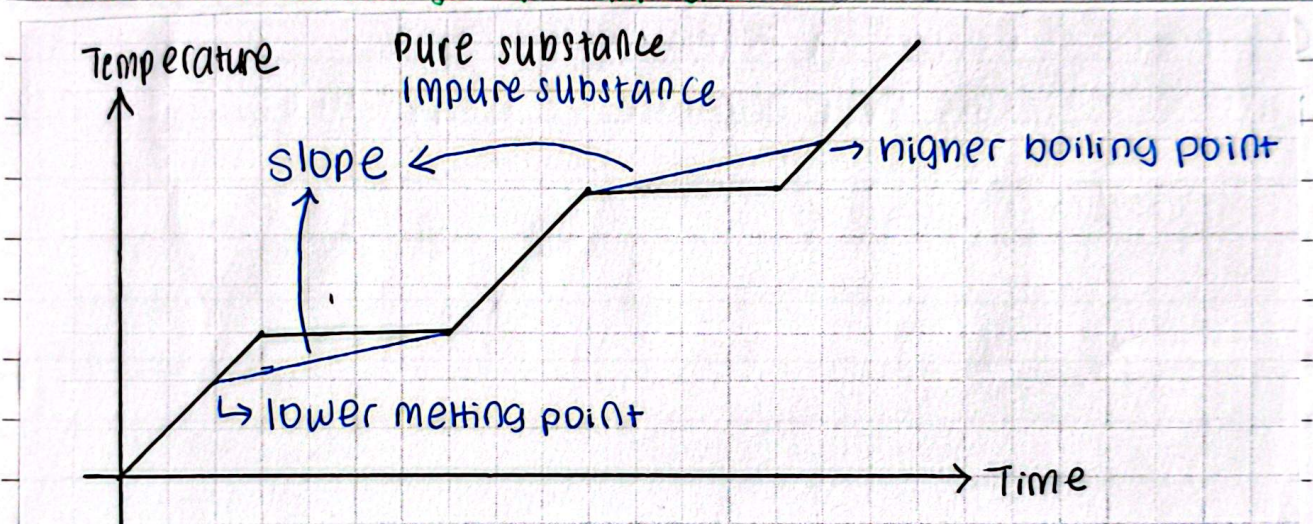
- When a pure substance is contaminated with impurities, a mixture is formed.
- Impurities affect melting & boiling points of a pure substance.

Effect of impurities on the melting point of substance

- Melting point is lower than the pure substance.
- Melts over a range of temperatures.

Effect of impurities on the boiling point of substance

- Boiling point is higher than the pure substance.
- Boils over a range of temperatures.



Compound vs Mixture

1. Separation

Compound → can only be separated into its elements by chemical reactions.

Mixture → can be separated into its components by physical methods. → e.g. filtration / separation technique

2. Formation

Compound → chemical change takes place when compound is formed

Mixture → No chemical change takes place when mixture is formed

3. Composition

Compound → Chemical composition is fixed, elements are combined in a fixed proportion by mass.

Mixture → Proportion of the constituents can vary.

4. Properties

Compound → Properties of a compound differ from those of its constituent elements.

Mixture → Properties of a mixture are the same as those of its constituent elements.

5. Melting and Boiling

Compound → Occurs at fixed temperature.

Mixture → Occurs over a range of temperatures.

Chapter 4: Chemical Bonding

Stable electronic configuration of noble gases

- Noble gases are found in group 18.

Properties of noble gases

- Noble gases are unreactive and stable.

↳ Their valence electron shells are completely filled with electrons.

- They either have two valence electrons (duplet electronic config) or eight valence electrons (octet electronic config)

Forming ions

- Atoms achieve stable noble gas configuration by:

1. losing electrons

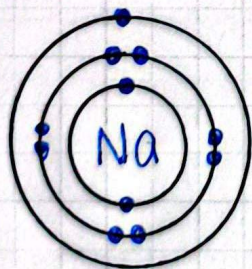
2. gaining electrons

3. sharing electrons

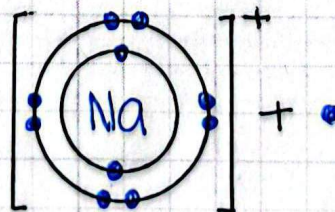
- When an atom gains or loses electrons, an ion is formed.
- An ion is a charged particle when an atom gains or loses electrons

Positive ions (cations)

- Cations are formed when atoms lose electrons.
- Elements in group 1 to 13 lose electrons to form cations with stable noble gas configuration. (i.e. metals = cations)



Sodium Atom
11 electrons
Na
2,8,1



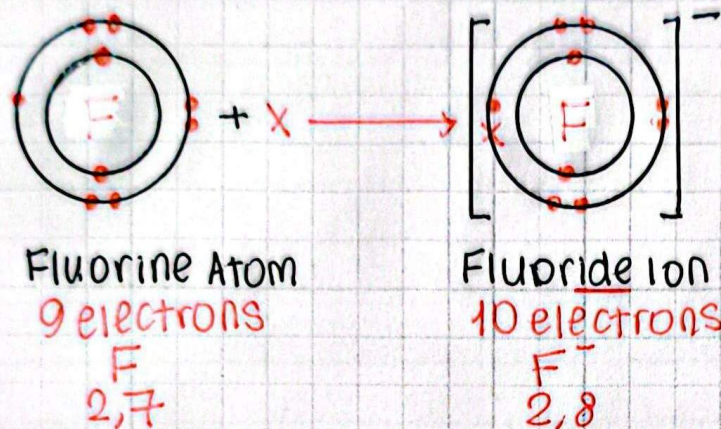
Sodium Ion
10 electrons
Na⁺
2,8

Example:
Sodium atom → ion

* No change in
protons &
neutrons

Negative Ions (anions)

- Anions are formed when atoms gain electrons.
- Elements in group 15 to 17 gain electrons to form anions with stable noble gas configuration. (i.e. non-metals = anions)



Example:
Fluorine atom → ion

- ★ For anions, the name of the non-metal has "ide" when they become ions
- ★ No change in protons & neutrons

Ionic Bonding: How to draw

Step 1: Are the elements metal or non-metal?

↳ Ionic bonding → Metal + Non-metal

Step 2: Write the electronic configurations

Step 3: Deduce Formula of ionic compound (Criss-cross)

Step 4: Draw

Example: Calcium and Oxygen

Step 1: Metal or non-metal?

Ca: Metal

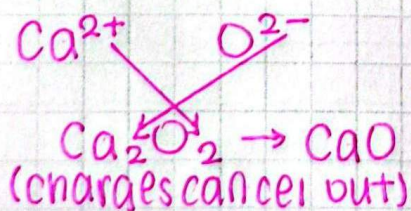
O: Non-metal

Step 2: Electronic config.

Ca: 2, 8, 8, 2 $\xrightarrow{\text{lose } 2e^-}$ [Ca]²⁺

O: 2, 6 $\xrightarrow{\text{gain } 2e^-}$ [O]²⁻

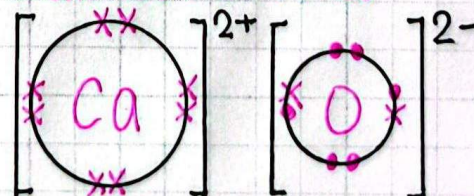
Step 3: Deduce formula



Chemical Formula of
Calcium oxide: CaO

Step 4: Draw

(valence electrons)



Legend:

- x → electron of Ca
- → electron of O

Ionic Compounds

- An ionic compound is a neutral substance that consists of ions of opposite charges held together by ionic bonds.
- Ionic compounds have no net charge.
- Ionic compounds/bonds are formed by a metal and a non-metal.

Covalent Bonding (How to draw)

Step 1: Are the elements metal or non-metal?

↳ Covalent bonding → Non-metal + Non-metal

Step 2: Write the electronic configurations

Step 3: Need — share — and deduce formula

Step 4: Draw

Example: Phosphorous and Fluorine

Step 1: Metal or non-metal Step 2: Electronic config.

P → Non-metal
F → Non-metal

P → 2, 8, 5
F → 2, 7

Step 4: Draw (valence electrons)

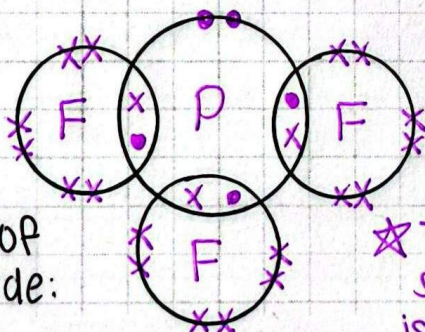
Step 3:

P: Need 3, share 3
F: Need 1, share 1

Cross-multiply:

P_1F_3
↓
 PF_3

Chemical Formula of
Phosphorous Fluoride:
 PF_3



Legend

x → electron of F
• → electron of P

★ The element sharing more e^- is in the middle.

Structural Formula (Covalent Bonding)

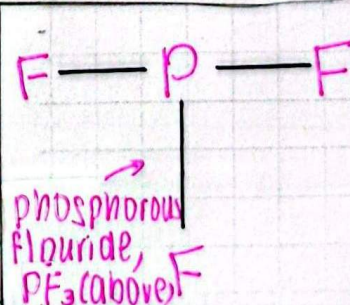
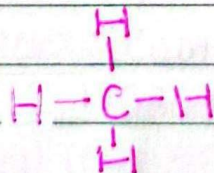
- For each pair of electrons shared, one line is drawn.

Examples:

$O=O$ $H-Cl$

$H-H$ $O=C=O$

$Cl-Cl$ $N \equiv N$



Covalent molecules/compounds

- A covalent molecule is a group of 2 or more atoms that share valence electrons.
- Covalent molecules/bonds are formed by non-metals only.
- Covalent bonds can be formed between:
 - atoms of the same element
 - atoms of different elements

Metallic Bonding

- Metal atoms lose valence electrons and become positively-charged ions.
- The metallic bond is the strong electrostatic force of attraction between positive metal ions and the 'sea' of delocalised electrons.
- Delocalised electrons refer to the electrons that have been lost from the valence shell of the atom and are free to move within the metal.

2 types of covalent bonding

- For each pair of electrons shared

one line is drawn

Examples:

